

Mad Lab : a game that only moves when you do

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Project Description:

For this project, I built an interactive browser-based game using JavaScript and the HTML Canvas API. The project evolved from my original concept of pixel-based animation controlled by the user, and expanded into a fully playable real-time game where the player's physical body is the controller. Rather than using a mouse or keyboard, the player stands in front of their webcam and uses their actual movements to interact with the game world. This is made possible by MediaPipe Pose, a Google AI model that detects 33 body landmarks in real time directly in the browser.

The game is called Mad Lab. The player takes on the role of a mad scientist defending a secret experiment , a glowing central orb, from waves of cartoon enemies trying to shut it down. The orb charges over time, and the player must protect it across three increasingly difficult rounds to win. The player physically hits enemies by moving their body into them, blocks incoming projectiles by stepping in front of them, leans left or right to dodge, and raises both arms above their head to activate a temporary shield around the orb. Every interaction is driven entirely by physical movement, derived from my original project concept. I chose to make this transition because it not only was more complex, but I've wanted to gain more experience in game design – I thought this would be a good experience to try it out. I am also really interested in AI, so I'm excited about actually prototyping with an AI model.

Minimum Viable Product (MVP)

- Real-time body tracking via webcam using MediaPipe Pose
- Player skeleton rendered on screen driven by body landmarks
- Enemies spawn and move toward the central orb
- Body collision destroys enemies
- Three round game loop with win and game over states

Target product

- Projectile system where enemies shoot at orb
- Lean dodge mechanic detected from hip movement over time
- Shield mechanic activated by raising both arms above shoulders
- Charge bar that fills over time and reacts to hits and damage
- Visual feedback for all interactions: bursts, flashes, etc

Ideal Product

- Cartoon scientist character drawn over the body skeleton
- Fully illustrated lab background environment
- Four distinct cartoon enemy types with unique behaviors
- Pose-based special moves

Development Roadmap

1. Set up HTML Canvas game loop and basic state management for rounds, lives, and charge
2. Integrate MediaPipe Pose – connect webcam feed, extract landmarks, convert to screen coordinates
3. Implement point to line segment collision detection
4. Build enemy spawn system with speed and frequency scaling per round
5. Add projectile system – enemies shoot at orb, player blocks with body
6. Detect shield pose and hip lean from landmark positions over time
7. Polish interactions, and refine design aspects; test across different user environments – lighting conditions/distance